

Schawlow's Ruler Notes and commentary—and a homework

Without re-deriving the basic equations given in handouts, shining the laser light on a set of rulings gives Bright constructive interference fringes for spots where

$$n\lambda = d(\cos(\alpha) - \cos(\beta_n)). \quad 1.$$

The ruling spacing is d , wavelength for HeNe, α is the incident angle with respect to ruler (not normal), and β_n is the angle of the n th fringe. I will also call the distance to the screen “ L ”---but is called x_0 in some of the papers.

If you make the small angle approximation (which is the same as expanding for small y_2), and convert the trig functions to Cartesian statements—then you get:

$$y(\text{approx.})_n^2 \sim (2\lambda L^2/d)n + y_0^2 \quad 2.$$

Where n is the interference order. The formula is approximate since we have used the small angle approx., which of course must break down at sufficiently large n . So the question of how large an n can I keep and analyze data in a simple manner needs to be looked at.

At present we can measure interference positions vs. n , and determine the wavelength by fitting equation 2 to a straight line, finding the slope and solving for the wavelength. The fitting might give an appropriate uncertainty in slope. That works fine as long as we don't violate small angle approx.

If instead of making the small angle approximation we solve for y^2 exactly (after putting things in Cartesian terms), then we get something like.

$$y_n^2 = L^2 \left[\frac{1}{\left(\frac{n\lambda}{d} - \frac{1}{\text{const}} \right)^2} - 1 \right] \quad 3$$

To figure out the constant you will need to actually derive this yourself starting from equation 1. So we have two different functions for y_n^2 . If we make a model using a tool like excel (yuck), origin, python, matlab—or your choice---then we can see how far off these two different approaches are. In other words when does equation 2 give significantly different values than equation 3 (the latter being exact—and not a straight line in n)?

You should use the wavelength for a HeNe—assuming no uncertainty.

$L = 7.00\text{m} \pm 0.07\text{m}$

$d = 0.500\text{mm}$ (no uncertainty—for ease)

$y_0 = 25\text{cm} \pm 1\text{cm}$.

1. Setup a model to generate a column (array) of values for $n=0$ to ~ 100 or so for y_n^2
2. Plot the two (together on one graph)---but put appropriate error bars on the points for the linear model.
3. Determine what value of n for which the approximate model and the exact model disagree by more than that points (the n th point) error bar.
4. What is your value of n ?
5. What value of n should you keep in order to ensure a good fit for your data?
6. For whatever $n(\text{max})$ you decided upon---your data for the exact model and use $n=0$ and $n=$ your $n(\text{max})$ and treat that like two points forming a line---use that to determine a slope and wavelength. How big of an effect does this have on the determination of the wavelength?
7. When analyzing this lab you have choices to make---you could do a non-linear fit to the exact function if you so desire (tell me before you make such a call)---OR you could use the linear fit approximation as instructed, but you must (MUST) be sufficiently careful.